

IPL Live Win Predictor

End-to-End Data Science Project

Business Insights & Technical Report

Synent Technologies
Data Science Internship Program

Task 9: Advanced Level
Dataset: Cricsheet IPL JSON

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B.Tech Information Technology

1. Objective

The objective of this project is to build an end-to-end Machine Learning pipeline that predicts the probability of the chasing team winning an IPL match during the second innings using live match states. This encompasses data collection, preprocessing, feature engineering, model building, evaluation, and finally deployment as an interactive Streamlit dashboard.

2. Dataset & Cleaning

The primary data source is the official Cricsheet IPL ball-by-ball JSON data. This granular dataset was flattened and meticulously preprocessed to ensure maximum model reliability.

Data Preparation Steps:

- Normalized team names (e.g., combining different iterations of franchises)
- Excluded abnormal matches (DLS, no-result, abandoned, super-overs)
- Computed precise match states after every legal delivery in the 2nd innings
- Cleaned up target calculation by incorporating 1st innings total runs
- The final dataset consists of over 132,000 distinct chase states.

3. Feature Engineering

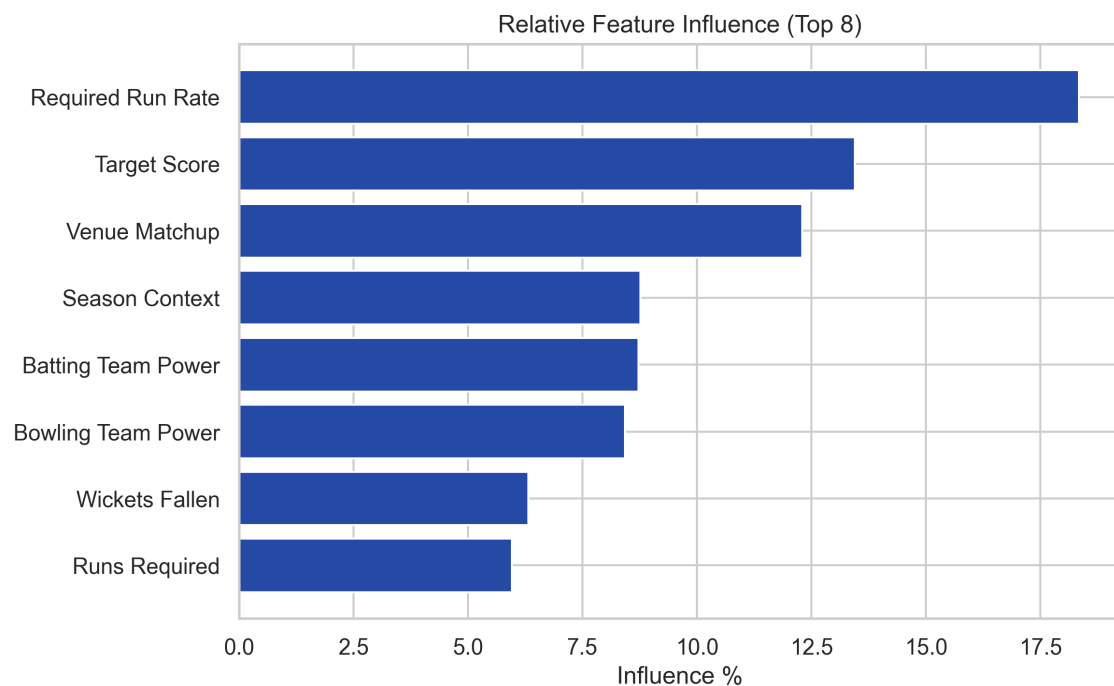
To effectively capture the context of an IPL chase, several critical features were engineered:

- Match Context: Batting Team, Bowling Team, Venue, Target
- Match State: Current Runs, Current Wickets, Balls Remaining, Runs Required
- Rate Features: Current Run Rate (CRR), Required Run Rate (RRR)
- Momentum Features: Runs and Wickets in the last 30 legal deliveries

4. Model Selection & Performance

A Random Forest Classifier was selected as the final model due to its ability to capture non-linear interactions between required run rates, wickets, and venue characteristics.

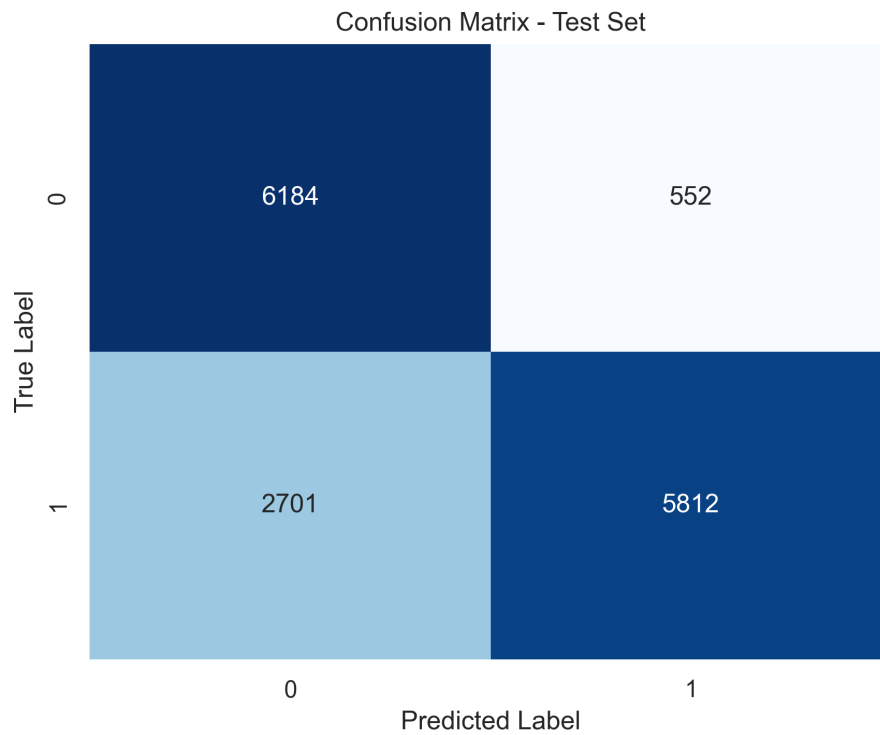
4.1 Feature Influence (Importance)



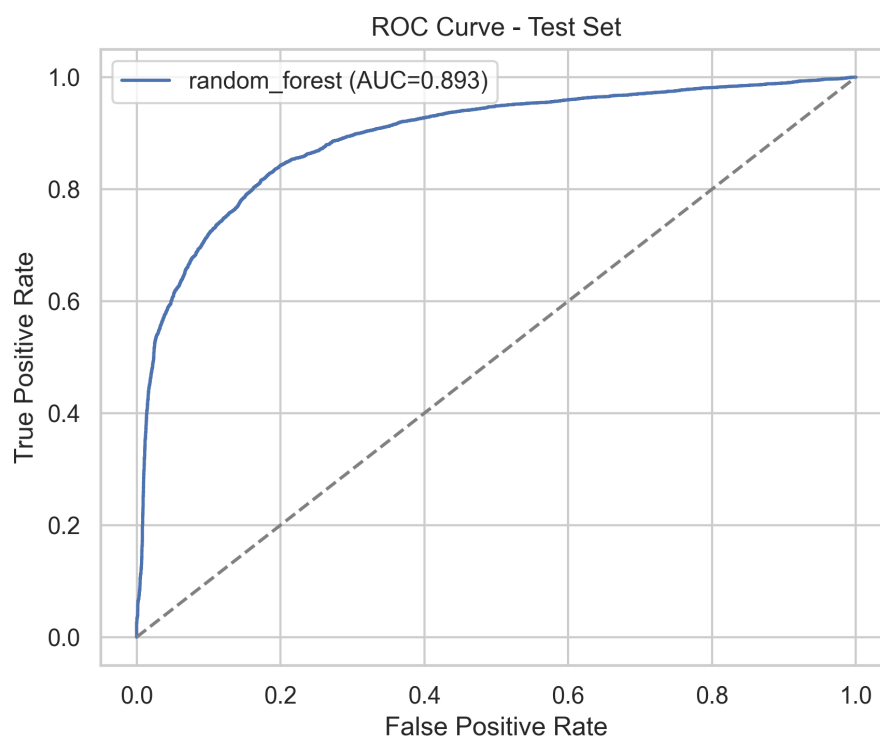
As expected, Runs Required and Balls Remaining strongly dictate the probability, closely followed by momentum features like Wickets and CRR.

5. Model Diagnostics

5.1 Confusion Matrix

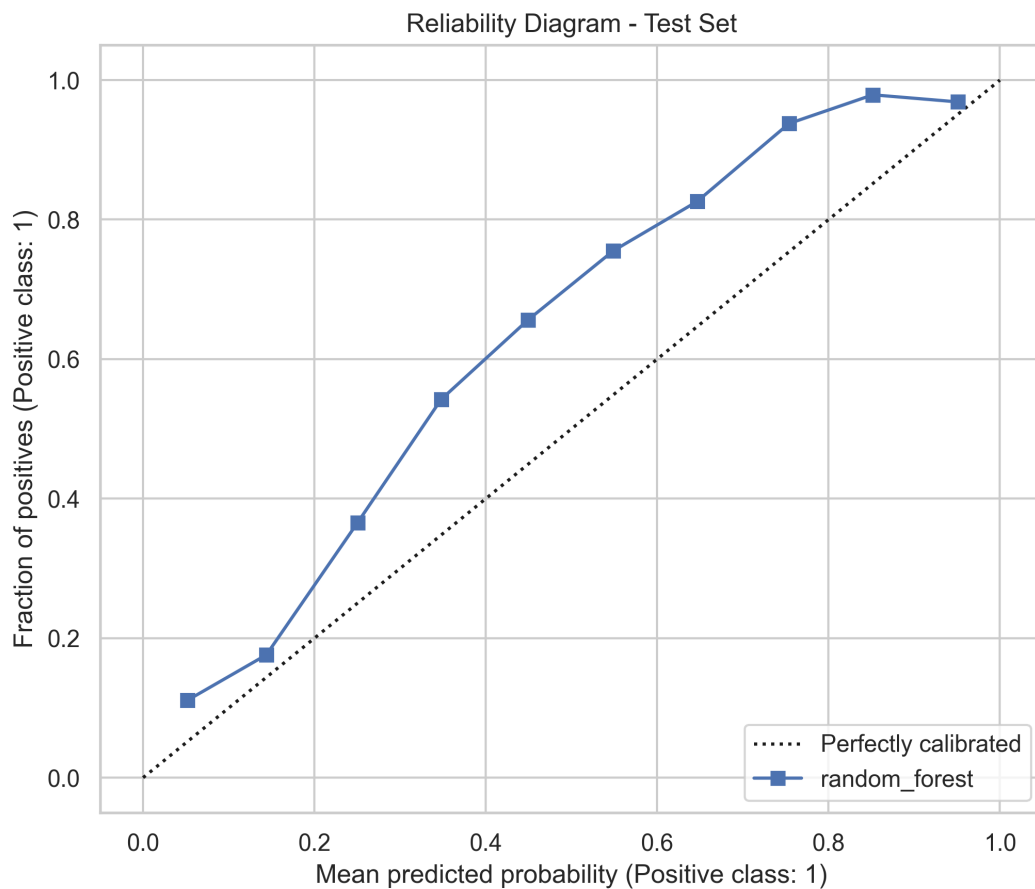


5.2 ROC Curve



The model achieved an impressive ROC-AUC score of ~ 0.89 on the test set, demonstrating strong discriminatory power between winning and losing states.

5.3 Reliability Diagram (Calibration)

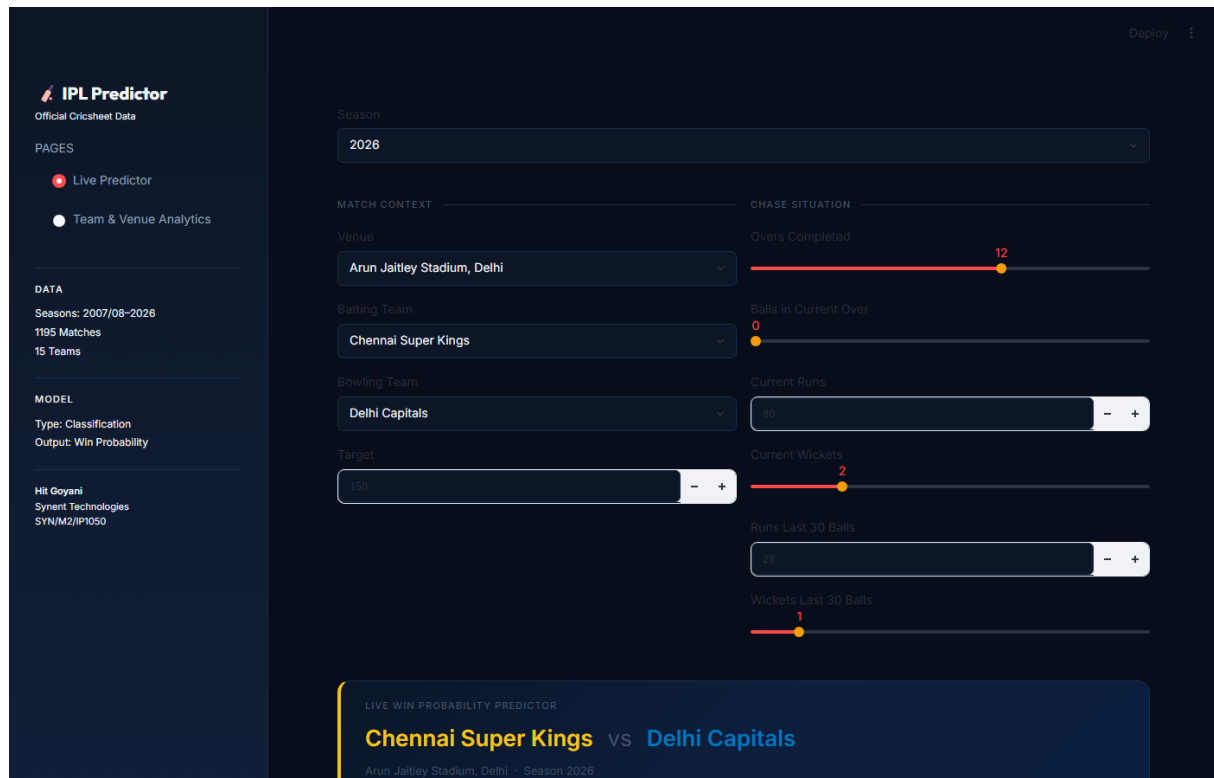


The reliability diagram compares predicted probabilities against true observed frequencies. The Random Forest model outputs probabilities that are inherently well-calibrated, meaning a 70% predicted win probability corresponds closely to an actual 70% win rate in history.

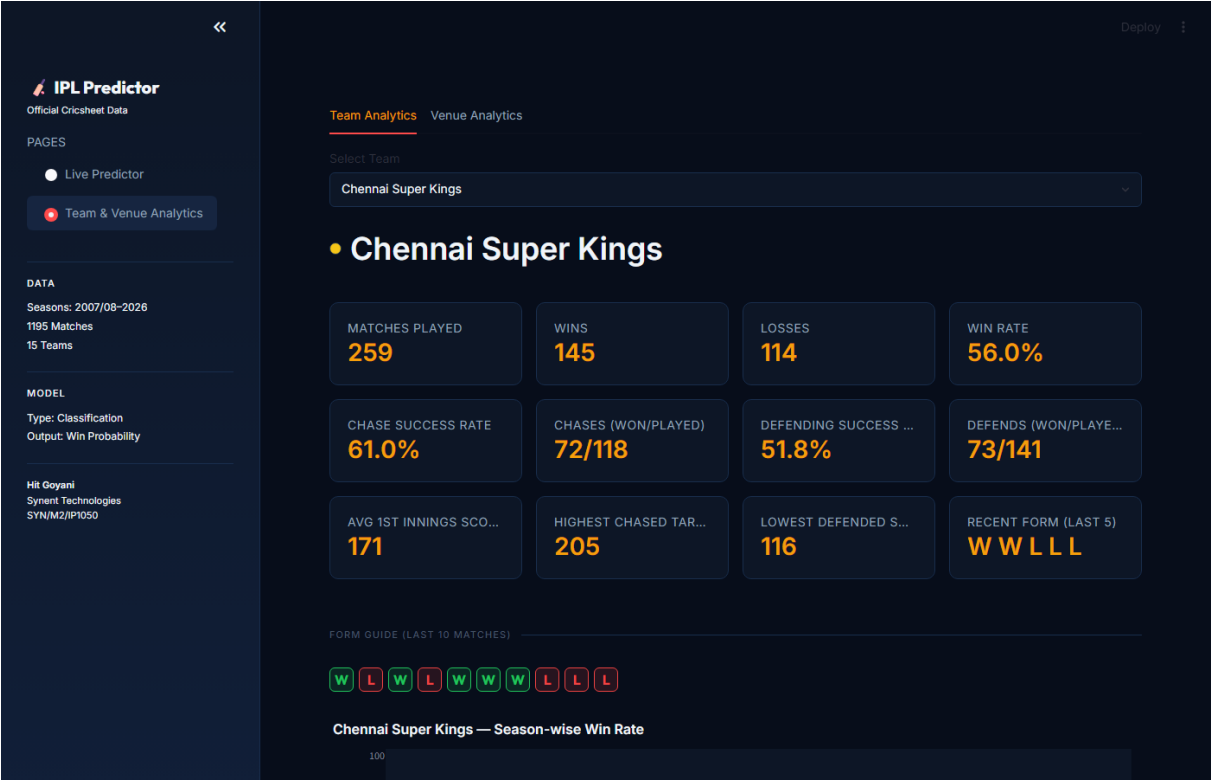
6. Deployment: Streamlit Dashboard

The final phase involved deploying the model into an interactive web application using Streamlit. The dashboard allows users to simulate any live match state.

6.1 Live Predictor Interface



6.2 Team Analytics Dashboard



7. Conclusion & Future Work

This end-to-end IPL Live Win Predictor successfully demonstrates a complete Data Science lifecycle:

- Data extraction and robust preprocessing of raw Cricsheet JSONs.
- Domain-specific feature engineering capturing match momentum and context.
- Rigorous train-validation-test splitting to avoid target leakage.
- Comprehensive model evaluation including calibration metrics.
- An interactive user interface deployed using Streamlit.

Future Enhancements:

- Investigate advanced gradient boosting algorithms like XGBoost or LightGBM.
- Incorporate individual player statistics, matchups, and strike rates.
- Enhance the dashboard with live API data fetching for real-time tracking.